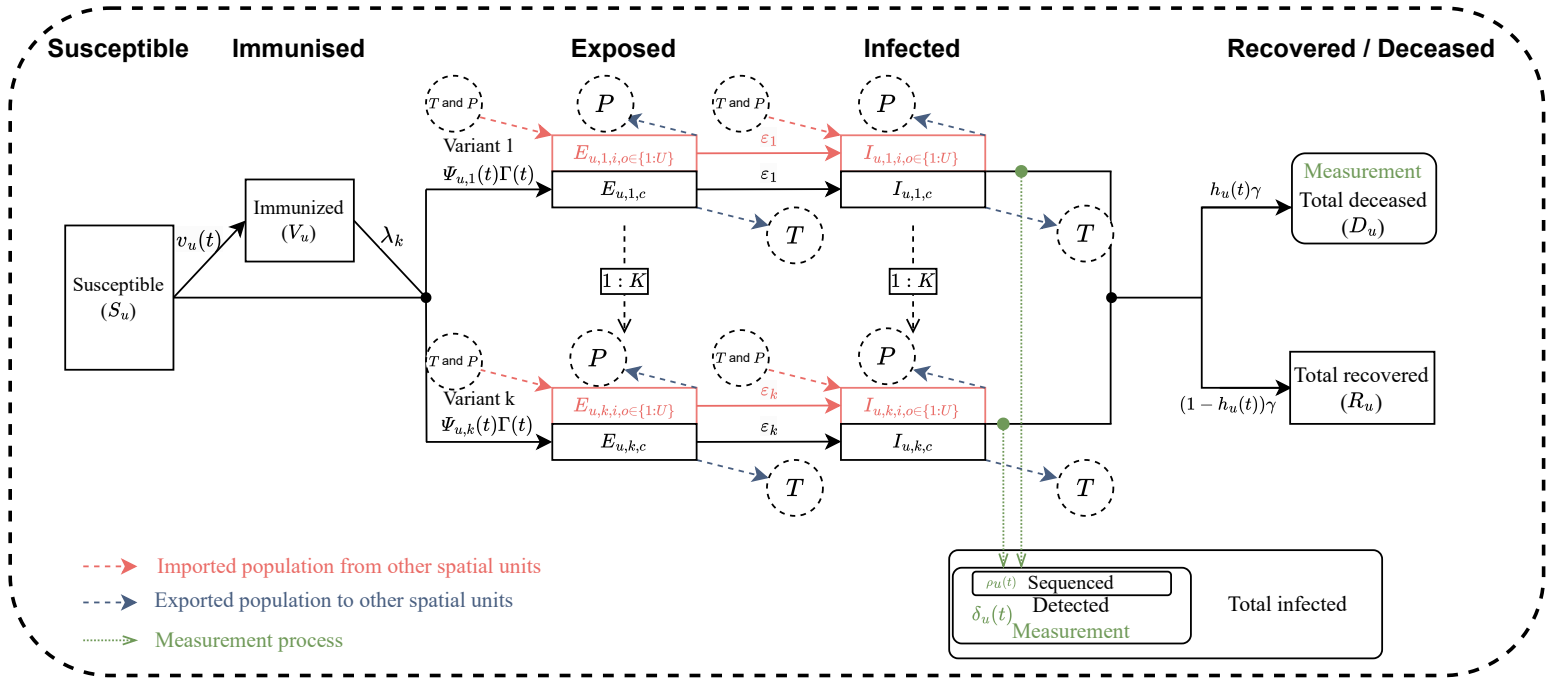
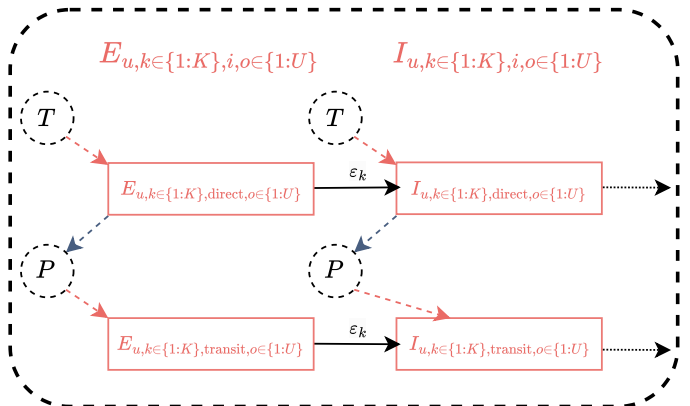


a



b



c

$$\Psi_{u,k}(t) = \eta_k \beta_u \frac{\tilde{I}_{u,k}(t)}{N_u(t)}$$

$$\tilde{I}_{u,k}(t) = \zeta_u(t) I_{u,k,i}(t) + I_{u,k,c}(t)$$

$$v_u(t) : \text{Vaccination rate at time } t \text{ for unit } u$$

$$\eta_k : \text{Relative increase/decrease in infection rate due to strain } k$$

$$\lambda_k : \text{Relative decrease in infection rate due to immunity for strain } k$$

$$\zeta_u(t) : \text{Change in transmissibility due to level } n \text{ travel control policy at time } t \text{ for unit } u$$

$$\varepsilon_k : \text{Inverse of the latent/incubation period for strain } k$$

$$\gamma : \text{Rate at which infected individuals become recovered or deceased}$$

$$h_u(t) : \text{Infection-to-fatality ratio at time } t \text{ in unit } u$$

$$\delta_u(t) : \text{Infection-to-diagnostic ratio at time } t \text{ in unit } u$$

$$\rho_u(t) : \text{Diagnostic-to-sequencing ratio at time } t \text{ in unit } u$$

β_u : Natural transmissibility in u
 $N_u(t)$: Population size at time t for unit u
 $\Gamma(t)$: Gamma white noise with intensity σ